

Tahsin Touheed Ivan

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EDUCATION

University of Dhaka

B.Sc. in Electrical and Electronic Engineering (currently 4th year)

Dhaka, Bangladesh

Sept 2022 – Dec 2026(expected)

CGPA: 3.73/4.00

Relevant Coursework: Analog and Digital Electronics, Electromechanical Energy Conversion, Solid State Physics, Communications Systems, Digital Signal Processing, Electronic Devices, Electromagnetic Theory and Antennas, Optoelectronics, Materials Science, Industrial and Biomedical Instrumentation, Microprocessor and Interfacing etc.

Notre Dame College

Higher Secondary Certificate (Science)

Dhaka, Bangladesh

Aug. 2019 – July 2021

GPA: 5.00/5.00

RESEARCH

Materials Science Research Laboratory DU

Jan 2026 – Present

Undergraduate Researcher

Supervisor: *Imtiaz Ahmed*

- Developing a structured literature dataset of porous carbon electrodes and applying machine learning to uncover quantitative relationships between structural properties such as — BET surface area, pore size distribution, nitrogen/oxygen heteroatom content — and specific capacitance, with the goal of identifying high-performance material configurations through data-driven screening.
- Experimentally validating model-shortlisted configurations through synthesis and characterization, covering the full pipeline from precursor carbonization and chemical activation through structural characterization (BET, Raman, XPS) and electrochemical testing (CV, GCD, EIS) — closing the loop between computational prediction and physical material behavior.

LEADERSHIP

IEEE SIGHT Student Branch DU

Feb 2025 – Present

Chair

April 2026 – Present

- Supervising development of technology-based solutions targeting real-world societal challenges, while driving collaborations with IEEE chapters to expand technical and community impact.

Program Coordinator

Feb 2025 – March 2026

- Coordinated and organized the “*Machine Learning Fundamentals*” workshop for undergraduate students; managed logistics, content planning, and participant engagement.
- Conducted hands-on workshop in the ‘*IEEE Day Outreach Programme on Microcontrollers and Their Utilization*’, jointly organized by the *IEEE Student Branch University of Dhaka*, *IEEE EDS SBC DU*, and *IEEE WIE AG SB DU* at *Nawab Faizunnesa Govt Girls High School, Cumilla*, 50+ high school students.

IEEE Photonics Society Student Branch DU

Feb 2025 – March 2026

Event Coordinator

- Organized technical workshops, seminars, and community outreach programs.

Optics and Photonics Club

Jan 2024 – April 2025

Vice President

- Worked in projects, seminars and events hand-in-hand with OPTICA.
- Gave talks, and project showcases(LFR, Holography etc.) in the outreach program ‘*Envisioning Harmony Between Human and Nature*’ at *St. Joseph Higher Secondary School, Dhaka*, organized in collaborations with the *Optica Bangladesh Student Chapter, University of Dhaka*, and *Scintilla Science Club*.

Project: Drone (Funded by FABLAB DU) | Python, Arduino, ESP32

2024

- Co-developed FabLab DU's first departmental drone (ESP32 quadcopter UAV, UoD EEE) within USD 66 (approx.) — integrated F450 Quad-X frame, A2212 1000KV BLDC motors, 30A ESCs, MPU6050 IMU, and FlySky FS-i6 RC system on an ESP32 DevKit V1 running ESP-FC/Betaflight firmware; achieved 2.8:1 TWR, $\pm 5^\circ$ hover stability, 15 m altitude, and 13 min flight time; platform architected for extensibility with GPS (Neo-6M), barometric altimeter (BMP280), FPV camera, LiDAR obstacle avoidance, and DSHOT ESC upgrade — all without a flight controller hardware change.
- Gained end-to-end embedded flight systems experience — configured ESP32–ESC PWM communication (1000–2000 μ s) and PPM/SBUS RC integration; implemented and hand-tuned cascaded PID control (rate + angle loops) across roll, pitch, and yaw; diagnosed and resolved ESC desync, IMU yaw drift, and vibration instability; developed practical understanding of centre-of-gravity balancing and its effect on flight dynamics; explored ESP32's dual-core architecture and onboard Wi-Fi as a basis for future telemetry, autonomous navigation, and sensor fusion — validating commodity microcontrollers as viable low-cost alternatives to STM32-based commercial flight controllers.

ESP32 Retro Gaming Console | ESP32

2026

- Built a handheld retro gaming console prototype using an ESP32, a 0.96 SSD1306 OLED, an analog joystick, and a passive buzzer — running three original arcade games (Car, Snake, Pong) with a scrollable menu, idle sleep animation, and per-game high scores saved to flash.
- Primary learning focus was ESP32 real-time firmware design — millis()-based finite-state machines, ADC noise filtering with hysteresis, and debounce logic for reliable input. Secondary focus covered peripheral integration: I²C display rendering with U8g2, LEDC PWM audio, and NVS for non-volatile storage across deep-sleep wake cycles.

Pocket Reader — ESP32 Handheld E-Book Reader | ESP32

2026

- Built a self-contained handheld e-book reader prototype using an ESP32, a 0.96 SSD-1306 OLED, and an analog joystick, serving a mobile-friendly web portal directly from the microcontroller for book uploads and Wi-Fi management — no cloud, no external server. Designed for extensibility: an e-ink display would add paper-like readability, and a physical keyboard would unlock a distraction-free writing mode.
- Primary learning focus was ESP32 networking — Wi-Fi station and access point modes, mDNS, and hosting a REST HTTP server on-device — alongside NVS, LittleFS, and deep-sleep power cycles. BLE hardware was a deliberate choice to preserve the option of a future phone-sync feature without requiring shared Wi-Fi.

LASER HARP | Arduino

2023

- An academic project from sophomore year, aim was to play two octaves span of musical notes using leds'.

COMPETENCIES**Programming Languages:** Python, C/C++, x86 Assembly, LaTeX**Softwares:** Verilog (RTL design), PSPICE, Proteus, MATLAB, Simulink, AutoCAD, EasyEDA**Languages:** English(IELTS band- 7), Bangla(native)**Soft skills:** Public Speaking, Team Leadership, Mentoring, Event Management, Media Coordination and Direction